Arunabha Majumder



Mechanical Engineer

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PROFILE

Mechanical engineer who builds and tests mechanisms for medical and assistive devices. I design from first principles, prototype fast with 3D printing, test on physical setups and on people, and change the design when it fails. My last device, a shoulder exoskeleton, won the Best Paper Award at IFToMM ISRM 2026.

EXPERIENCE

PhD Researcher, Robotics (project VIEXO)
Aalborg University, Denmark | *2023 - Present*

- Designed a hybrid shoulder exoskeleton from first principles: a variable stiffness mechanism with a worm gear for spring pretension, working in parallel with a 6 Nm motor. It uses more than 25% less motor energy, which means longer battery life.
- Optimised the mechanism with a genetic algorithm (diameter, spring stiffness, pretension force), built the CAD and drawings in SolidWorks, 3D printed prototypes and then had the final parts manufactured.
- Integrated the motor (actuation), encoder and torque/force sensors; measured the stiffness on a physical test setup to characterise performance.
- Failure and manufacturability: the first clutch kept breaking and was hard to make, so I replaced it with a simple bolt-locking clutch (8 locking angles) that was easy to make and reliable.
- Designed and modelled a novel 5-bar planar parallel robot with variable stiffness mechanisms (VSM); estimated its stiffness with less than 10% error and quantified human arm impedance (6 participants).
- Collected EMG data from 12 participants with the Qualisys system; implemented an Arduino-based trigger to sync robot motion with Qualisys recordings.
- Taught university courses and supervised Bachelor's students (2023 - 2025).

MTech Researcher, Soft Exosuit (Pneumatic Muscles)
CSIR-CMERI, India | *2020 - 2022*

- Fabricated and assembled pneumatic artificial muscles for a soft exosuit; designed the outer PID loop of a neural network-based controller in MATLAB/Simulink and implemented it in LabVIEW to drive the pneumatic muscle (0.3 - 0.78% error, IEEE CONECCT 2022).

Bachelor's Capstone and Student Teams
Siddaganga Institute of Technology, India | *2017 - 2019*

- Designed and built an arm exoskeleton and exo glove for a paralysed arm: CAD, machining, assembly and testing (TATA Technologies Prayatna Innovation Fund grant; Best Major Project Award).
- Built a Go-Kart (2017) and an Efficycle hybrid vehicle (2018) with student teams.

SKILLS

Mechanical design: mechanisms from first principles, machine elements (worm gears, springs, clutches), tolerances, actuation (electric motors, pneumatic muscles), DFM and DFA

CAD and prototyping: SolidWorks (CSWA), technical drawings, 3D printing (Bambu Lab, Ultimaker Cura), machining and hands-on assembly; Fusion 360 (basic, last used 2019)

Testing: torque/force sensors, encoders, Qualisys motion capture and EMG, Arduino, MATLAB/Simulink, LabVIEW, Python (basic)

Art and graphic design: hand sketching of concepts, painting, technical illustration; Adobe Illustrator, Photoshop, Procreate

EDUCATION

PhD, Mechatronics and Robotics, Aalborg University, Denmark | 2023 - present (research complete)

MTech, Mechatronics, AcSIR - CSIR-CMERI, India | 2020 - 2022

BTech, Mechanical Engineering, Siddaganga Institute of Technology, India | 2015 - 2019

Senior Diploma in Fine Arts (Painting), Bangiya Sangeet Parishad | 2008, First Division with Distinction

LANGUAGES AND REFERENCES

English (C1), Bengali (native), Hindi (spoken), French (willing to learn)

References: Prof. Shaoping Bai (sbh@mp.aau.dk); Assoc. Prof. Lars Rosgaard Jensen; Assoc. Prof. Anderson de Souza Castelo Oliveira; all Department of Materials and Production, Aalborg University